

Importing necessary libraries

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
from plotly.offline import init_notebook_mode, iplot
init_notebook_mode connected=True)
```

Loading dataset

```
In [2]: df = pd.read_csv("open_tickets.csv")
```

Exploratory Data Analysis(EDA)

```
In [3]: df.head()
```

Out[3]:

	Ticket ID	Customer Name	Customer Email	Customer Age	Customer Gender	Product Purchased	Date Purch
0	6	Rebecca Fleming	sheenasmith@example.com	53	Male	Microsoft Office	2020-
1	7	Jacqueline Wright	donaldkeith@example.org	24	Other	Microsoft Surface	2020-
2	8	Denise Lee	joelwilliams@example.com	23	Male	Philips Hue Lights	2020-
3	19	Wendy Davis	brenda20@example.net	19	Male	LG Washing Machine	2021-
4	23	Stephanie Nelson DVM	ljohnson@example.org	54	Female	Xbox	2020-

5 rows × 24 columns



```
In [4]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2819 entries, 0 to 2818
Data columns (total 24 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Ticket ID                            2819 non-null   int64
1   Customer Name                        2819 non-null   object
2   Customer Email                      2819 non-null   object
3   Customer Age                        2819 non-null   int64
4   Customer Gender                    2819 non-null   object
5   Product Purchased                  2819 non-null   object
6   Date of Purchase                   2819 non-null   object
7   Ticket Type                        2819 non-null   object
8   Ticket Subject                     2819 non-null   object
9   Ticket Description                  2819 non-null   object
10  Ticket Status                      2819 non-null   object
11  Resolution                         0 non-null      float64
12  Ticket Priority                    2819 non-null   object
13  Ticket Channel                    2819 non-null   object
14  First Response Time                0 non-null      float64
15  Time to Resolution                 0 non-null      float64
16  Customer Satisfaction Rating       0 non-null      float64
17  Age Group                         2819 non-null   object
18  Is Resolved                       2819 non-null   bool
19  Cleaned_Description               2819 non-null   object
20  Response Delay (in hours)          0 non-null      float64
21  Resolution Time (in hours)         0 non-null      float64
22  Customer_Response_Pending_Days    0 non-null      float64
23  Auto_Close                       2819 non-null   bool
dtypes: bool(2), float64(7), int64(2), object(13)
memory usage: 490.2+ KB
```

```
In [5]: sunburst_data = df.groupby(['Age Group', 'Ticket Type', 'Ticket Subject']).size(

fig = px.sunburst(
    sunburst_data,
    path=['Age Group', 'Ticket Type', 'Ticket Subject'],
    values='Count',
    title="Sunburst Chart: Ticket Distribution by Age Group, Type, and Subject"
)

fig.update_layout(
    paper_bgcolor="#E6E6FA",
)

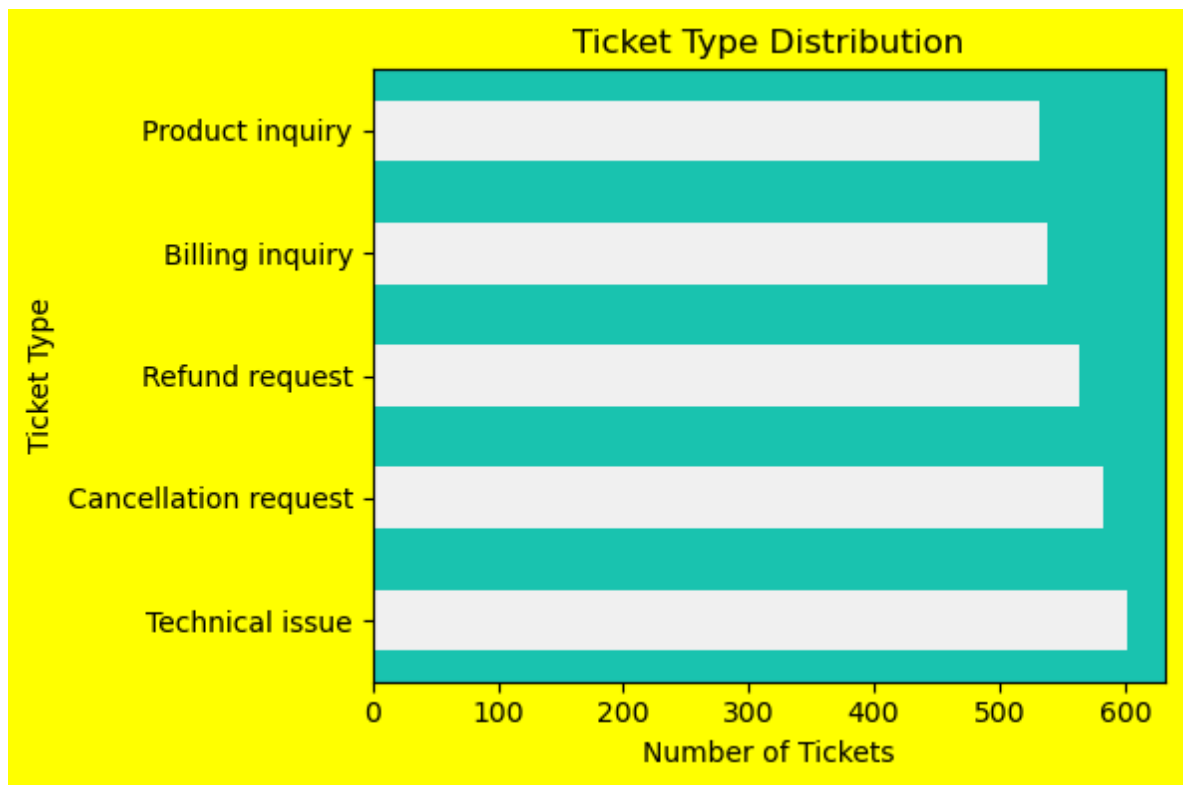
iplot(fig)
```

Sunburst Chart: Ticket Distribution by Age Group, Type, and Status

```
In [6]: # Counting the number of each ticket type
ticket_counts = df['Ticket Type'].value_counts()

# Plotting horizontal bar chart
plt.figure(figsize=(6,4),facecolor= 'yellow')
ax= ticket_counts.plot(kind='barh',color="#F0F0F0")
ax.set_facecolor("#1CC4AF")

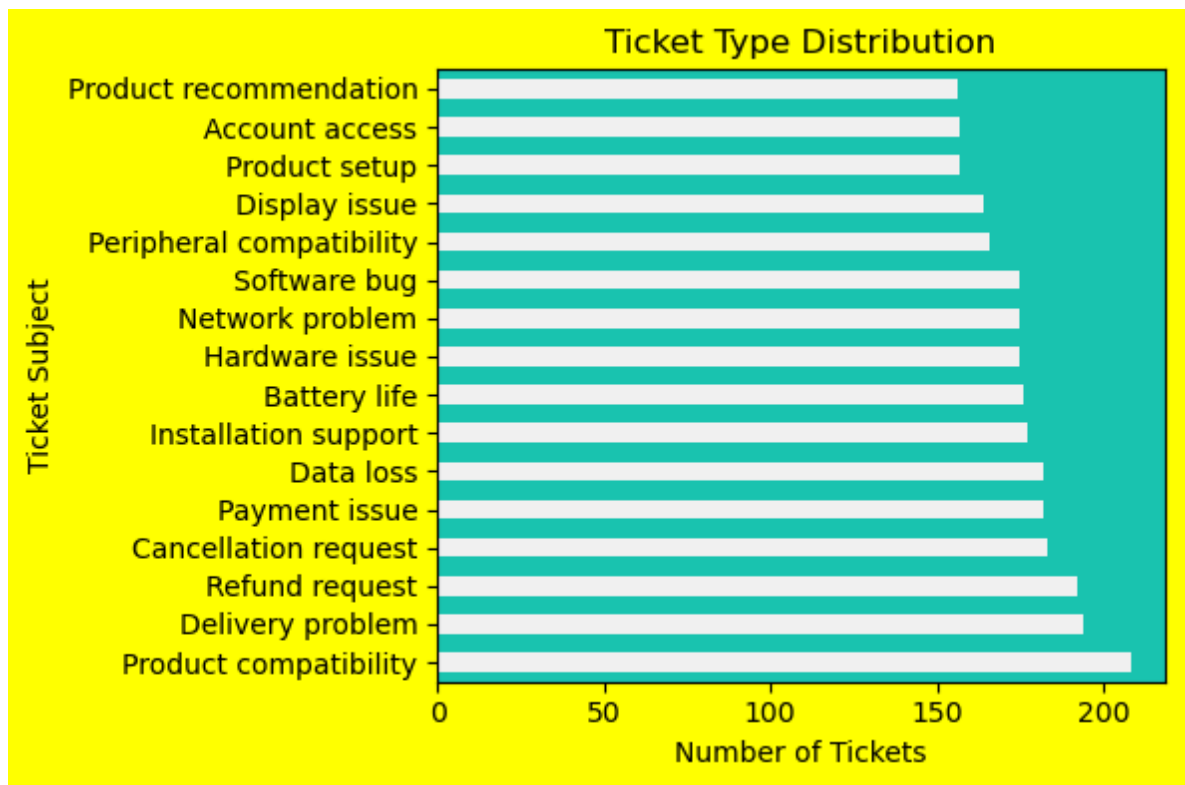
# plot
plt.title('Ticket Type Distribution')
plt.xlabel('Number of Tickets')
plt.ylabel('Ticket Type')
plt.tight_layout()
plt.show()
```



```
In [7]: # Counting the number of each ticket subject
ticket_counts = df['Ticket Subject'].value_counts()

# Plotting horizontal bar chart
plt.figure(figsize=(6,4),facecolor= 'yellow')
ax = ticket_counts.plot(kind='barh',color="#F0F0F0")
ax.set_facecolor("#1CC4AF")

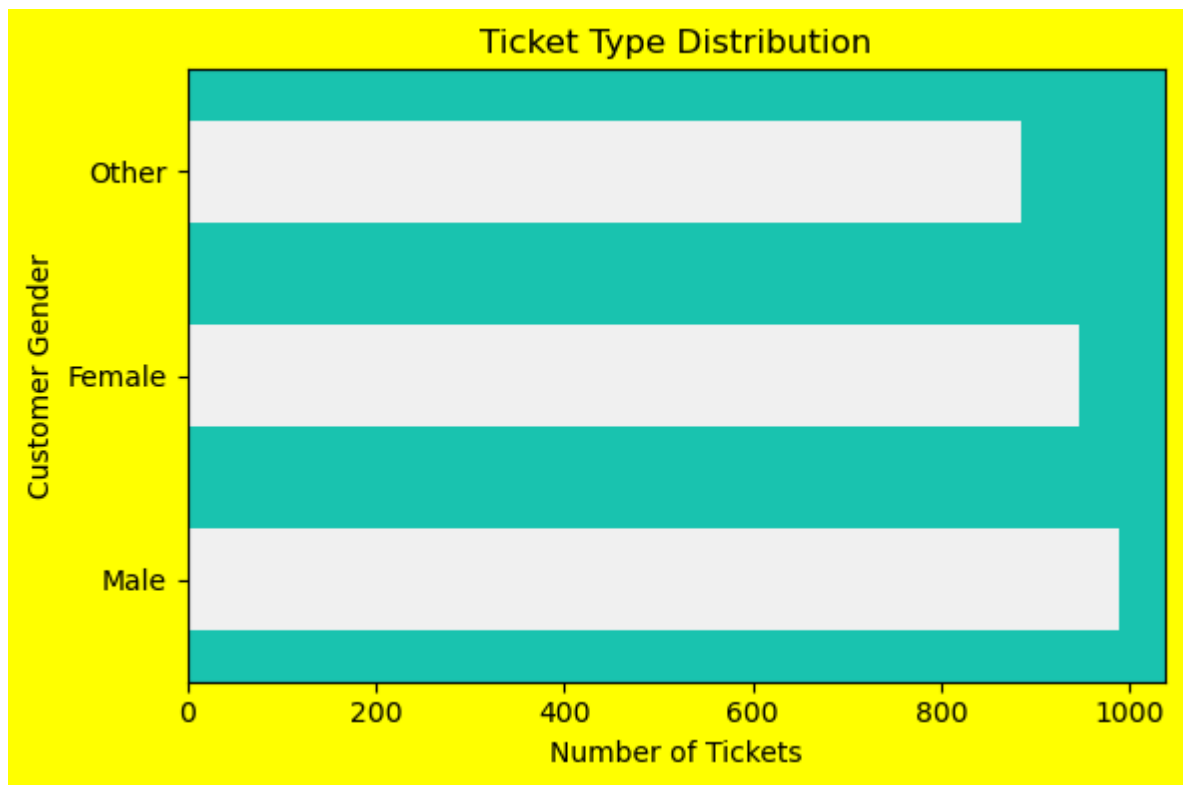
# plot
plt.title('Ticket Type Distribution')
plt.xlabel('Number of Tickets')
plt.ylabel('Ticket Subject')
plt.tight_layout()
plt.show()
```



```
In [8]: # Counting the number of Gender category
ticket_counts = df['Customer Gender'].value_counts()

# Plotting horizontal bar chart
plt.figure(figsize=(6,4),facecolor= 'yellow')
ax = ticket_counts.plot(kind='barh',color="#F0F0F0")
ax.set_facecolor("#1CC4AF")

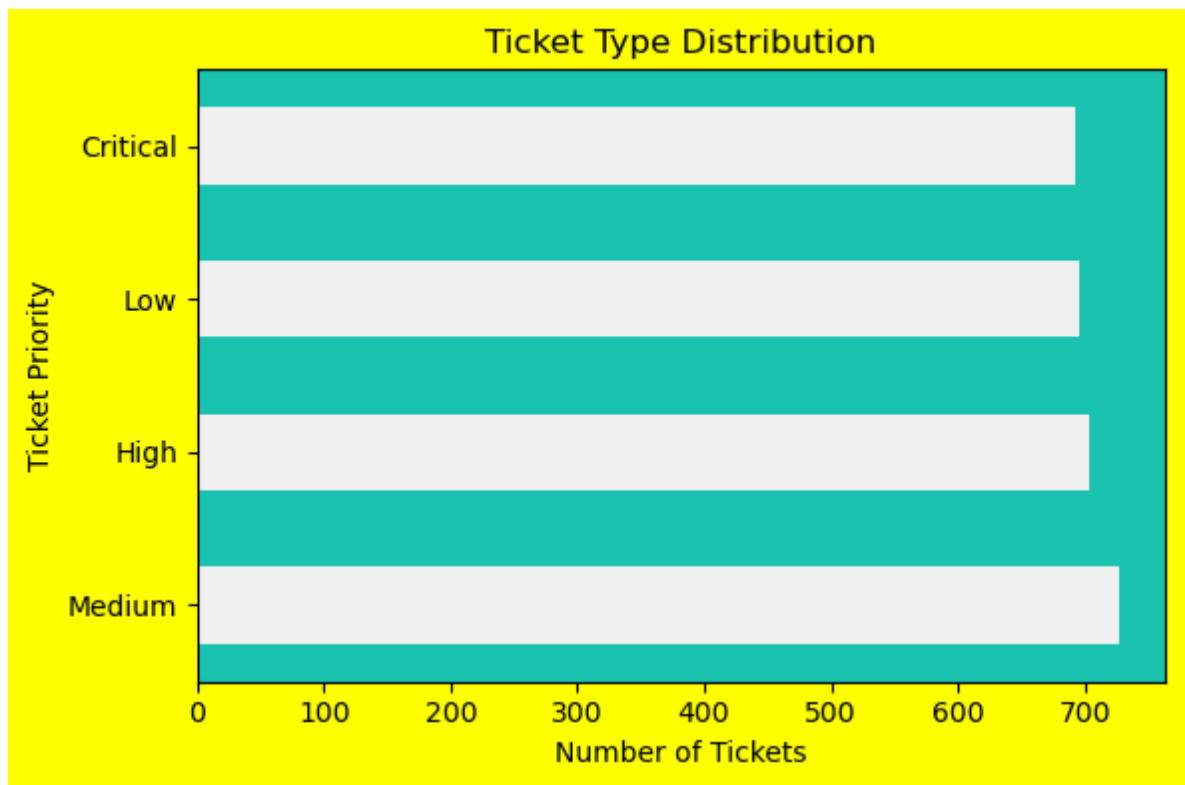
# plot
plt.title('Ticket Type Distribution')
plt.xlabel('Number of Tickets')
plt.ylabel('Customer Gender')
plt.tight_layout()
plt.show()
```



```
In [9]: # Counting the frequency of each ticket priority
ticket_counts = df['Ticket Priority'].value_counts()

# Plotting horizontal bar chart
plt.figure(figsize=(6,4),facecolor= 'yellow')
ax = ticket_counts.plot(kind='barh',color="#F0F0F0")
ax.set_facecolor("#1CC4AF")

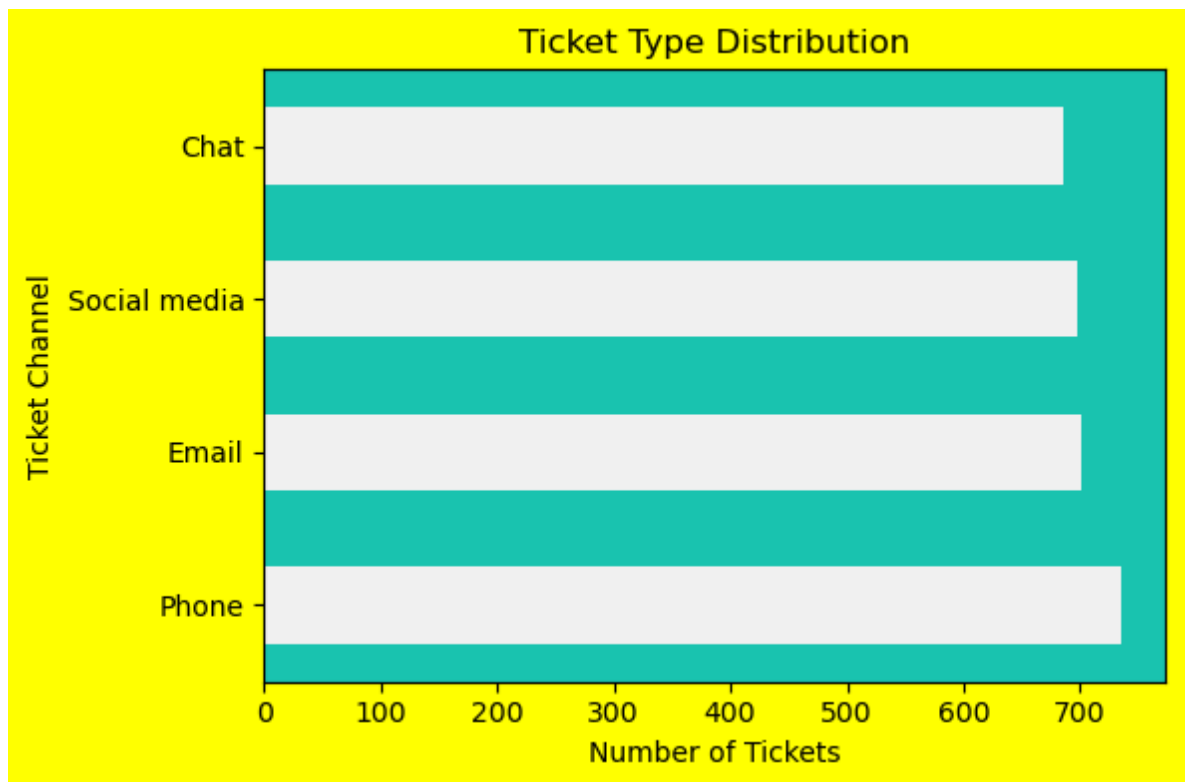
# plot
plt.title('Ticket Type Distribution')
plt.xlabel('Number of Tickets')
plt.ylabel('Ticket Priority')
plt.tight_layout()
plt.show()
```



```
In [10]: # Calculating the distribution of ticket channel
ticket_counts = df['Ticket Channel'].value_counts()

# Plotting horizontal bar chart
plt.figure(figsize=(6,4),facecolor= 'yellow')
ax = ticket_counts.plot(kind='barh',color="#F0F0F0")
ax.set_facecolor("#1CC4AF")

# plot
plt.title('Ticket Type Distribution')
plt.xlabel('Number of Tickets')
plt.ylabel('Ticket Channel')
plt.tight_layout()
plt.show()
```



Summary of Unresolved Ticket Data (No Initial Response)

What we found:

In the **open_tickets.csv** file, several tickets were found with **no initial response** from the support team. These tickets remain open and unattended, which can lead to poor customer experience and dissatisfaction.

Why this matters:

When customers don't receive a response at all, it creates a negative impression of the support process. Delayed or absent replies reduce trust and may result in loss of customer loyalty.

What we can do:

- **Set up an alert system** to flag all tickets that remain unanswered for more than a few hours or a day, so they can be escalated quickly.
- **Assign higher priority** to such tickets and regularly review them in team meetings to ensure no ticket goes unnoticed.
- **Train support staff** to respond promptly, even if only to acknowledge the ticket and let the customer know their issue is being reviewed.
- **Use a chatbot** to send immediate auto-replies or collect basic information from the customer while they wait. This builds confidence and keeps them engaged.
- **Monitor response times** and introduce a KPI (Key Performance Indicator) for "First Response Time" to drive accountability.

- **Review ticket assignment logic** to ensure open tickets aren't left unassigned or stuck in the system.

Overall Impact:

By focusing on unresolved tickets and taking proactive steps to respond quickly, we can significantly improve customer satisfaction, prevent escalation of issues, and maintain a positive brand image.

In []: